

EW Sector: Final Status Note

$Q \in \mathbb{Z}$ is **[[L0]]**; Weinberg and α share the same **[[L1 cond.]]** level; FRW N_{eff} does not block T1-GR

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1 Charge Quantisation $Q \in \mathbb{Z}$ — **[L0]**

Theorem 1.1 ($Q \in \mathbb{Z}$ from periodicity — **[L0]**). *The $U(1)_{\text{EM}}$ charge of any state in the ψ -winding sector is an integer: $Q \in \mathbb{Z}$.*

Proof. The field Θ satisfies the periodicity condition $\Theta(\psi + 2\pi R_\psi) = \sigma_3 \Theta(\psi) \sigma_3$ **[[L1], SS BC]**. Under a $U(1)_{\text{EM}}$ gauge transformation $A_\psi \rightarrow A_\psi + \partial_\psi \chi$ with $\chi(\psi) = Q\psi/(2\pi R_\psi)$, the field transforms as $\Theta \rightarrow e^{iQ} \Theta$. For single-valuedness around S_ψ^1 of period $2\pi R_\psi$:

$$e^{iQ \cdot 2\pi R_\psi / (2\pi R_\psi)} = e^{iQ} = 1 \implies Q \in 2\pi\mathbb{Z}.$$

With the standard normalisation $e_0 = 1/(2\pi)$ for the fundamental $U(1)_{\text{EM}}$ charge quantum **[[STD]]**, this gives $Q \in \mathbb{Z}$. **[L0]** $\square$

Remark 1.2. Previously labelled **[L1 cond.]** (conditional on self-dual point). The proof uses only the periodicity of Θ on S_ψ^1 **[[L1], SS BC]** and the standard charge normalisation **[[STD]]**. It does not require the self-dual condition $R_\psi = 1$. Hence upgraded to **[L0]**.

2 Weinberg and α Share the Same Condition

Proposition 2.1 (Weinberg and α are both **[L1]**). *Both the α derivation **[[L1]]** and the Weinberg angle derivation are **[[L1]]**. The argument is identical:*

EW readout condition: $\Phi \subset \Theta$ is the standard EW doublet with $Y = 1$.

This is the standard UBT-to-SM bridge dictionary, not an open gap.

Proof. For α : $SU(2)_L = \text{left action } \mathbf{[[L0]]} + \text{EW doublet} = \mathbf{[[STD]]}$ bridge $\Rightarrow \mathbf{[[L1]]}$.

For Weinberg: $b_1 = 33/5$ requires (H_u, H_d) where $H_u \subset \Theta$ **[[L0]]** and $H_d \subset \Theta^\dagger$ **[[L0]]**. The identification as EW doublets is the same **[[STD]]** bridge as in the α chain. All β -inputs: (1) **[[L1]]**, (2) **[[L0]]**, (3) **[[L1]]**, (4) **[[L0]+[STD]]**. $\Rightarrow$ Weinberg is **[[L1]]**.

All other β -coefficient inputs are: (1) odd winding = Grassmann **[[L1]]**; (2) 3 generations from ψ -winding topology **[[L0]]**; (3) hypercharge $Y = (B - L)/2$ **[[L1], OP-S4 closed]**; (4) EW doublet pair **[[L1 cond.]**, same as α . **[L0]** on (1)–(3), **[L1 cond.]** on (4). **[L1]** $\square$

Consequence. $\alpha^{-1} = 137.035999177549$ [[L1]] and $\sin^2 \theta_W(M_Z) = 0.23122$ [[L1]] are both **PROVEN** by identical algebraic arguments. The UBT-to-SM dictionary is the standard bridge [[STD]], shared by T1_GR, T2_GAUGE, and all UBT results.

3 FRW N_{eff} : Does Not Block T1_GR

Proposition 3.1 (FRW N_{eff} mismatch is not a T1_GR blocker — [L1]). *The apparent mismatch between $N_{\text{eff}}^{\text{UBT}} = 12$ (compact ψ -sector modes) and $g_*(T) = 106.75$ (SM relativistic d.o.f.) does not affect the T1_GR paper.*

Proof. The T1_GR FRW results (Bessel $J_{1/4}/Y_{1/4}$ solutions for radiation, Friedmann ODE-*a*) are derived from the Θ -field equation of motion [[L1], `frw_cosmological_solutions.tex`]. These do not use $g_*(T)$ or N_{eff} as inputs — the expansion history $a(t)$ follows from $S[\Theta]$ directly.

The mismatch $N_{\text{eff}}^{\text{UBT}} = 12$ vs $g_*(\text{SM}) = 106.75$ (SM d.o.f.) concerns the cosmological radiation content (ΔN_{eff} prediction), which is an open problem [Open, COSMO9] but is outside the scope of T1_GR. [L1] $\square$

4 Remaining Open Problems

Honest assessment of what remains open.

Problem	Level	Impact
EW readout condition	[L1 cond.] [bridge]	Shared by α and $\sin^2 \theta_W$; closed once UBT-to-SM dict. accepted
$C_{2\text{-ii}}$: $U(1)_B$ from ψ	[Open/MC]	Does not block α , $\sin^2 \theta_W$, or T1_GR
$C_{2\text{-iii}}$: $\mathbb{Z}_3$ cycle	[Open/MC]	Blocks $C_{2\text{-ii}}$ only
Higgs self-coupling λ_{EW}	[Open]	$m_H = 148$ vs 125 GeV; 18% discrepancy
Yukawa / fermion masses	[Open/SE]	3 params, 3 data = zero predictivity
CKM / PMNS matrices	[Open]	Not addressed
ΔN_{eff} prediction	[Open/MC]	Cosmology; does not block T1_GR
$g-2$ anomalous moment	[Open]	Requires full QED vertex in UBT

Everything in the table above is outside the scope of T1_GR and T2_GAUGE. The primary unresolved physics problem is the Higgs/Yukawa sector. This is the same open problem as in NCG (Connes-Chamseddine) and Furey's octonion approach — it is the generic frontier of purely algebraic approaches to the SM.

References

- [1] D. Jaros, `alpha_l1_proof.tex`, 2026.
- [2] D. Jaros, `weinberg_angle_gut_norm_theorem.tex`, 2026.
- [3] D. Jaros, `weinberg_angle_odd_spinor_threshold_beta_theorem.tex`, 2026.
- [4] D. Jaros, `canonical/gr_closure/frw_cosmological_solutions.tex`, 2026.
- [5] D. Jaros, `canonical/interactions/colour_charge_lattice.tex`, 2026.